BỘ GIÁO DỤC VÀ ĐÀO TẠO
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ KỸ THUẬT TP. HCM
KHOA ĐIỆN – ĐIỆN TỬ



KỸ THUẬT DFT VÀ KIỂM THỬ

Career Opportunities in

IC Design at Marvell Vietnam

GVHD: ThS NGUYỄN VĂN THÀNH LỘC

NHÓM : 01

Marvell Vietnam · Ho Chi Minh City & Da Nang

Career Opportunities in
IC Design at Marvell Vietnam

From ASIC design flow to engineering roles, skills, compensation and a student-ready roadmap.

Lê Quốc Khánh - 24161276

Overview & introduction

Nguyễn Bửu Phúc-24161356

ASIC flow · RTL · DV

Nguyễn Hữu Đức - 24161221

DFT · PD · AMS · skills

Phạm Triệu Khang Duy - 24161204

Compensation · roadmap · Q&A

Marvell Vietnam · Ho Chi Minh City & Da Nang

LÊ QUỐC KHÁNH

Marvell Overview
& Vietnam R&D Hub

Why Vietnam matters in Marvell’s global engineering network—and what technologies are being built here.



COMPANY OVERVIEW

LÊ QUỐC KHÁNH

Marvell Technology & Vietnam R&D Hub

Data Infrastructure Semiconductor

Marvell Technology

Semiconductor solutions for data centers, cloud, AI, networking, carrier infrastructure and enterprise storage.

- Data Center
- Cloud & AI
- Networking / 5G
- Storage

Vietnam engineering footprint

3rd largest global R&D hub

500+ engineers in Vietnam

Ho Chi Minh City

Da Nang

Vietnam teams contribute to core engineering—not only support functions.

Marvell Technology & Vietnam R&D Hub

TECHNOLOGY PORTFOLIO

LÊ QUỐC KHÁNH

What Does Marvell Design?

Custom Silicon

Custom ASIC & SoC

Purpose-built silicon for cloud, AI and data infrastructure workloads.

Switching

High-Speed Networking

Ethernet switching and network infrastructure for scale-up and scale-out data-center fabrics.

Interconnect

PAM4 / SerDes / Optical / PCIe & CXL

High-speed SerDes, optical/electrical interfaces and PCIe technologies for extreme bandwidth.

Storage

Storage Controllers

HDD, SSD and enterprise-storage controllers plus data-infrastructure accelerators.

ASIC DESIGN FLOW

NGUYỄN BỬU PHÚC

Where Each Career Path Fits

01
Specification
Architecture, interfaces, PPA targets

→

02
RTL + DV
Verilog/SystemVerilog, UVM, coverage

→

03
Synthesis + DFT
Netlist,constraints, scan, MBIST, ATPG

→

04
PD + SIGNOFF
Floorplan · PnR · CTS · STA · timing closure

→

05
Tape-out
Signoff and GDSII handoff

→

06
Validation
Board bring-up, lab measurements, silicon debug

Marvell Vietnam’s engineering roles map across nearly the full IC lifecycle—from architecture and RTL to physical implementation and silicon validation.

EARLY-CAREER ENTRY POINT

NGUYỄN BỬU PHÚC

Marvell Vietnam Internship 2026

66

internship positions

in the HCMC hiring snapshot

Target: Engineering students graduating in 2026–2027
Start: June 2026
Duration: typically 3–6 months

Digital Design	RTL implementation & synthesis-aware design	13
Analog Design	High-speed analog / mixed-signal circuits	13
Physical Design	Place & route, CTS, timing & signoff	10
Analog Layout	Custom layout, DRC/LVS, parasitics	6
DV · DFT · STA · SW/FW · HW	Verification, testability, timing and system roles	Other

FRONT-END CAREERS

NGUYỄN BỬU PHÚC

Digital RTL Design & Design Verification

Digital / RTL Design

Build the logic

Translate architecture into synthesizable RTL and drive blocks toward power, performance and area targets.

- Verilog
- SystemVerilog
- Digital Logic
- FSM
- Computer Architecture
- Synthesis
- Timing constraints

Typical collaboration: Architect ↔ RTL Designer ↔ Verification ↔ Physical Design

Design Verification

Prove the logic works

Develop verification plans, UVM environments, automated tests and functional coverage before silicon is built.

- SystemVerilog
- Regression
- Assertions
- Coverage
- Python/Perl
- UVM
- Linux

Mindset: systematic debugging, corner cases, automation and completeness.

SPECIALIZED CAREER TRACKS

NGUYỄN HỮU ĐỨC

DFT · Physical Design · Analog / Mixed-Signal

DFT ENGINEER

Make silicon testable

Add test structures so manufactured chips can be screened efficiently.

- Scan Chain
- MBIST
- ATPG
- Boundary Scan

PHYSICAL DESIGN / STA

Turn netlist into silicon

Build the physical implementation and close timing for signoff.

- Floorplan
- Placement
- CTS
- Routing
- Timing closure

ANALOG / MIXED-SIGNAL

Design transistor-level IP

Create and verify high-speed analog and mixed-signal circuits.

- SerDes
- PLL
- ADC / DAC
- SPICE / Spectre

NGUYỄN HỮU ĐỨC

Required Skills Matrix for IC Jobs

Track	HDL / UVM	DFT / Test	PnR / STA	SPICE / AMS	EDA Tools	Linux + Scripting
RTL Design	**Strong**	Basic	Basic	—	VCS / Xcelium	**Yes**
Verification	**Strong**	Basic	—	—	VCS / Xcelium	**Yes**
DFT	Basic	**Strong**	Medium	—	Tessent / DFT tools	**Yes**
Physical Design	Basic	Basic	**Strong**	—	Innovus / ICC2 / PT	**Yes**
Analog / Layout	—	—	Medium	**Strong**	Virtuoso / Spectre	**Yes**

Common baseline across tracks: technical English · Linux · scripting mindset · teamwork · problem solving

PHẠM TRIỆU KHANG DUY

Compensation, Benefits & Student Roadmap

Compensation & benefits

Internship allowance	**Competitive monthly support**
13th-month salary	**For full-time employees**
Performance bonus	**Variable by role / performance**
RSU	**Restricted Stock Units**
Health insurance	**Employee benefit package**

Avoid quoting a fixed salary unless you have a current, role-specific official source.

Student preparation roadmap

01 Foundation

Digital logic · analog basics · computer architecture

02 Practice

Verilog/SystemVerilog projects · GitHub portfolio

03 Tools

Linux · Python/TCL · simulation / EDA exposure

04 Communication

Technical English · teamwork · interview practice

PHẠM TRIỆU KHANG DUY

Choose a track. Build depth.

Then prove it with projects.

Marvell Vietnam offers career paths across RTL Design, Verification, DFT, Physical Design, Analog/Mixed-Signal and silicon-facing roles.

- RTL / Digital
- Verification
- DFT
- Physical / STA
- Analog / AMS

For students, the goal is not to master every domain at once—it is to select a direction, strengthen the fundamentals and build evidence through real projects.

MARVELL

Kết Luận

Qua bài thuyết trình, chúng ta đã có cái nhìn tổng quan về Marvell Vietnam, quy trình thiết kế ASIC và các vị trí nghề nghiệp chính trong lĩnh vực vi mạch như RTL Design, Design Verification, DFT, Physical Design/STA và Analog/Mixed-Signal.

Mỗi vị trí yêu cầu kiến thức và công cụ khác nhau, nhưng đều cần nền tảng về điện tử – vi mạch, Linux, kỹ năng lập trình/scripting, tiếng Anh kỹ thuật, làm việc nhóm và giải quyết vấn đề.

Đối với sinh viên, điều quan trọng là chọn một hướng phù hợp, xây dựng kiến thức nền tảng, thực hành qua các project thực tế và từng bước làm quen với các công cụ EDA. Đây sẽ là nền tảng để chuẩn bị tốt hơn cho cơ hội thực tập và làm việc trong ngành thiết kế vi mạch tại Marvell cũng như các công ty bán dẫn khác.

Cảm ơn sự lắng nghe của thầy và các bạn!